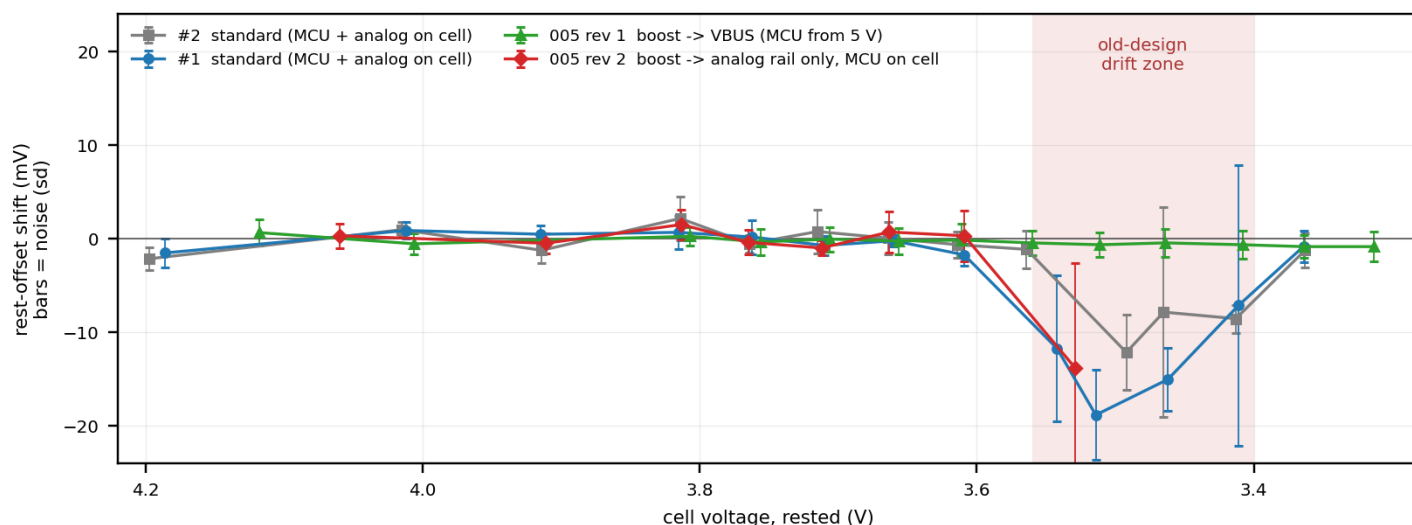


BlinkyHawk battery-range studies — Oct 1–3, 2026

Question: do the open/closed and voltage readings stay trustworthy as the cell runs down, and does moving the analog/MCU supplies onto a 5 V boost fix the low-battery drift? **Method** (per level): DL3021 load drains the cell to the target and rests it; Giga meter watches the sense node; then a battery-mode auto-tune with USB cut by jig K8 — DC sweep ± 1.2 V + AC, loads SHORT/10K/150K/1M/10.6M/OPEN, verify, deep-sleep wake check (150K must wake, 1M must not). Script: scratch *batt_study*.py* on top of the suite's Runner/BatteryBench.

Units and results

Unit / board	Power architecture	Range	Result
#2 SN 20260928_002	standard: MCU (XIAO LDO) + analog rail from cell	4.20–3.36 V	Stable to 3.56 V. Rest offset shifts $\sim$ 11 mV, noisier at 3.49–3.41 V; recovers by 3.36 V. Deep-sleep 150K creeps to .209 (needed SLEEPTHR11 0.228).
#1 SN 20260928_001	standard (same as #2)	4.19–3.36 V	Matches #2 within $\sim$ 2 % on every load metric and on wake. Drift starts earlier (3.55 V) and is larger ($\sim$ 19 mV, up to 15 mV noise); verify fails 9/14 at 3.46 V. Recovers by 3.36 V.
005 rev 1 SN 20260722_005	5 V boost into XIAO VBUS $\rightarrow$ MCU LDO from 5 V; analog on 3.3 V	4.12–3.31 V	Rock-stable: offset within 1.5 mV, noise 1.2–1.7 mV, gain 81–83 mV/V to 3.31 V. But never sleeps on battery (VBUS never drops so USB never detaches), CHGINHIBIT had to be 0, back-fed VBUS, flaky USB re-attach.
005 rev 2 (same board, modified)	boost feeds only the battery rail (analog + LED via gates); MCU back on the cell	4.06–3.53 V	USB detach, sleep/wake, CHGINHIBIT=1 all fixed . Clean to 3.61 V; at 3.53 V the drift returns ($\sim$ 14 mV, 11 mV noise, +0.6 V reads $\sim$ 0). Below 3.5 V not run (jig failures).



Shift = rest reading (nothing on the leads) relative to the same unit's average above 3.7 V. Absolute rest: standard units $\approx$ -25 mV, rev 1 $\approx$ +12 mV, rev 2 $\approx$ -28 mV.

What it shows

- **The low-battery drift follows the MCU's 3.3 V supply, not the analog rail.** It appears at $\sim$ 3.55 V on every board whose XIAO LDO runs from the cell (#1, #2, rev 2) and is absent when the LDO runs from 5 V (rev 1). Sense-node pulse amplitude agrees: #1 flat at 1.69 V down to 3.60 V then sagging; rev 1 flat 1.62–1.68 V to 3.31 V.
- **Recommended architecture:** keep rev 2's USB isolation (boost never touches VBUS; charge/VBUS sense from the connector) **and** give the MCU a regulated 3.3 V to the end of the cell (e.g. buck-boost on the 3V3 rail).
- **Capacity:** moving the flat-battery cutoff from $\sim$ 3.55–3.60 V to $\leq$ 3.3 V adds $\approx$ +10–15 % usable capacity (slow-drain OCV estimate). Runtime also depends on the supply path: boost + LDO from 5 V is only $\sim$ 55–60 % efficient vs $\sim$ 89 % for LDO-from-cell — measure battery current before choosing.
- **Thresholds** (old-design analog scale, method 2 tail area): 150K max $\approx$.19–.20, 1M min $\approx$.24–.26 V $\cdot$ ms $\rightarrow$ THRESH11 $\approx$ 0.222, SLEEPTHR11 0.228. Rev 1's scale was half that $\rightarrow$ method 0 (|diff|), threshold 0.535.

Gotchas found along the way

- **Firmware:** closing the COM port while the board is writing hangs it (RA4M1 SerialUSB spin) — *GuardedSerial* fix written but not yet flashed. Sleep is refused while *if (Serial)* is true, which never clears if VBUS stays up. With CHGINHIBIT=0 a board can sleep on USB $\rightarrow$ COM opens fail; wake it with a SHORT first.
- **Rev 2:** the first reading after USB is pulled is often a false VOLTAGE (4 of $\sim$ 15 cuts) — STABLECOUNT probably hides it from the user; check.
- **Harness:** aligner took the first lone V as the marker (fixed in scratch: needs 3 in a row); first condition of a capture cycle sometimes reads the previous state; passes.csv/summary.csv accumulate across levels in one process.
- **Rig:** 5 jig hangs / 2 bus-wide USB drop-outs during battery captures with the boost boards, at relay switches — suspect jig 5 V/relay-coil brownout; give the jig its own powered hub. Reset button is not enough (ESP32 bridge stays stuck): replug. Don't kill scripts that hold the jig port. Large cells need $>$ 150 mA drains.

Where things are

TestBench/BlinkyHawkTestSuite/Runs: * *BattStudy_SN20260928_002* (Oct 1–2), * *BattStudy_SN20260928_001* + * *AutoTune_u1study_** (Oct 2), * *BattStudy_SN20260722_005* + * *u3study_** (rev 1; ignore 2026-10-03_0645 — clips shorted), * *SN20260722_005_rev2* + * *u4study_** (rev 2). Saved configs: #1 THRESH11 .222 / SLEEPTHR11 .228 / REFCENTER $\sim$.0255; 005 CHGINHIBIT 1 (its other saved values are from rev 1 and need re-tuning for rev 2). Unit rows in BlinkyHawk_Unified/blinkyhawk_units.csv.